

Advanced Java Lab

Week-2

Roll Number: 238W1A12A7

1. Insert, Update, Delete Using PreparedStatement

Code:

```
/*  
  
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change  
this license  
  
 */  
  
package com.mycompany.mavenproject2;  
  
import java.sql.*;  
  
public class Mavenproject2 {  
  
    public static void main(String[] args) {  
  
        String url = "jdbc:mysql://localhost:3306/mydb";  
        String user = "root";  
        String password = "";  
  
        try {  
            Class.forName("com.mysql.cj.jdbc.Driver");  
            Connection con = DriverManager.getConnection(url, user, password);  
            System.out.println("Connected!");  
  
            // INSERT -----  
            String insertSQL = "INSERT INTO Person (PersonID, FirstName, City) VALUES  
(?, ?, ?)";  
            PreparedStatement insertStmt = con.prepareStatement(insertSQL);
```

```

insertStmt.setInt(1, 101);
insertStmt.setString(2, "Rahul");
insertStmt.setString(3, "Hyderabad");
int insertCount = insertStmt.executeUpdate();
System.out.println(insertCount + " row inserted.");

// UPDATE -----
String updateSQL = "UPDATE Person SET City=? WHERE PersonID=?";
PreparedStatement updateStmt = con.prepareStatement(updateSQL);
updateStmt.setString(1, "Bangalore");
updateStmt.setInt(2, 101);
int updateCount = updateStmt.executeUpdate();
System.out.println(updateCount + " row updated.");

// DELETE -----
String deleteSQL = "DELETE FROM Person WHERE PersonID=?";
PreparedStatement deleteStmt = con.prepareStatement(deleteSQL);
deleteStmt.setInt(1, 101);
int deleteCount = deleteStmt.executeUpdate();
System.out.println(deleteCount + " row deleted.");

con.close();

} catch (Exception e) {
    System.out.println("Error : " + e.getMessage());
}
}
}

```

Output:

```

--- exec:3.1.0:exec (default-cli) @ mavenproject2 ---
Connected!
1 row inserted.
1 row updated.
1 row deleted.
-----
BUILD SUCCESS
-----
Total time: 3.593 s
Finished at: 2025-11-22T17:39:40+05:30
-----

```

2. Explore ResultSet navigation methods and retrieve data in tabular format.

Code:

```

/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change
 this license
 */

```

```
package com.mycompany.mavenproject2;
```

```
import java.sql.*;
```

```

public class Mavenproject2 {
    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/mydb";
        String user = "root";
        String password = "";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);

            Statement stmt = con.createStatement(

```

```

        ResultSet.TYPE_SCROLL_INSENSITIVE,
        ResultSet.CONCUR_READ_ONLY
    );

    ResultSet rs = stmt.executeQuery("SELECT PersonID, FirstName, City FROM
Person");

    // Print table header
    System.out.println(" -----");
    System.out.printf("%-10s %-15s %-15s\n", "ID", "Name", "City");
    System.out.println(" -----");

    // Move to first record
    if (rs.first()) {
        do {
            System.out.printf("%-10d %-15s %-15s\n",
                rs.getInt("PersonID"),
                rs.getString("FirstName"),
                rs.getString("City")
            );
        } while (rs.next());
    }

    System.out.println(" -----");

    // Move to last record
    if (rs.last()) {
        System.out.println("Last Record → " + rs.getString("FirstName"));
    }

```

```

// Move to first record

if (rs.first()) {

    System.out.println("First Record → " + rs.getString("FirstName"));

}

// Move to absolute row number 2

if (rs.absolute(2)) {

    System.out.println("2nd Row → " + rs.getString("FirstName"));

}

con.close();

} catch (Exception e) {

    System.out.println("Error : " + e);

}

}

}

```

Output:

```

--- exec:3.1.0:exec (default-cli) @ mavenproject2 ---
-----
ID      Name      City
-----
1       John      New York
2       Rahul     Hyderabad
3       Meera     Mumbai
4       David     London
-----
Last Record ? David
First Record ? John
2nd Row ? Rahul
-----
BUILD SUCCESS
-----
Total time: 4.238 s
Finished at: 2025-11-22T17:44:05+05:30
-----

```